

DSD Final Project

Pipelined RISC-V Processor Final Submission

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Group 18

Processor Overview

- Five-stage pipelined RISC-V core
- Forwarding and stall units for data hazards
- Branch flush and prediction path
- I-cache and D-cache with slow-memory interface
- Baseline RV32I required instructions
- Compressed instruction fetch/decode
- Multiplication instruction support
- Cache flush support for final memory check

Cache Sweep Research

Pattern	Blocks	Ways	Misses	Exp. delay ns
QSort	16	1	1861	99945
QSort	32	1	387	27225
QSort	32	2	289	19080
QSort	64	2	119	7020
Conv	16	1	617	33435
Conv	32	1	103	6075
Conv	32	2	106	5490
Conv	64	2	69	3105
LFSR-HIST	16	1	43337	1967490
LFSR-HIST	32	1	593	43200
LFSR-HIST	32	2	588	42795
LFSR-HIST	64	2	566	40455

Observation: larger caches reduce miss cost, but high associativity gave limited benefit versus timing/area cost.

Branch Predictor Sweep Research

Predictor on BrPred	Accuracy	Storage
Always not taken	37.97%	0 FF
Always taken	62.03%	0 FF
BTFNT	73.42%	0 predictor FF
Global 3-bit counter	68.35%	3 FF
4-entry gshare 2-bit	86.08%	8 FF + history
8-entry bimodal 2-bit	93.67%	16 FF

Pattern	Always NT	BTFNT	Global 3-bit
QSort	72.44%	72.44%	70.48%
Conv	63.58%	66.05%	63.58%
LFSR-HIST	85.46%	85.46%	85.44%

Final choice: BTFNT gives useful BrPred gain with very low hardware cost.

Synthesis Result

Metric	Value
Clock cycle	3.00 ns
Combinational area	278998.244294
Noncombinational area	204061.428898
Buf/Inv area	38512.308615
Total cell area	483059.673192
Total area	4308345.930912

Source: 02_SYN/Report/CHIP_syn.area

Final Score Summary

Item	Result
BrPred execution cycles	526 cycles
hasHazard execution cycles	2235 cycles
Compression simulation time	1990 ns
Mul simulation time	718 ns
QSort simulation time	316447 ns
Conv simulation time	58963 ns
LFSR-HIST simulation time	3250843 ns
Performance score	2.93e22

Extension: Branch Prediction

- Prediction path before EX resolution
- BTFNT-style direction from branch immediate sign
- Wrong-path instructions flushed on EX redirect

```
[INFO]: Testing with pattern <BrPred>
-----
START!!! Simulation Start .....
-----
=====
Reset ...
=====
sh: ../00_TESTBED/info/success: Permission denied
=====

\(^o^)/ CONGRATULATIONS!! The simulation result is PASS!!!
=====
$finish called from file "../00_TESTBED/testbench/Final_tb.v", line 269.
$finish at simulation time 1579033
```

Extension: Compressed Instructions

- 16-bit instruction length detection in IF
- Decompression into internal 32-bit instruction form
- Handles 32-bit instruction crossing word boundary

```
[INFO]: Testing with pattern <compression>
-----
START!!! Simulation Start .....
-----
=====
Reset ...
=====
sh: ../00_TESTBED/info/success: Permission denied
=====

\(^o^)/ CONGRATULATIONS!! The simulation result is PASS!!!
=====
$finish called from file "../00_TESTBED/testbench/Final_tb.v", line 269.
$finish at simulation time 1990033
```

Extension: Multiplication

- Multi-cycle multiplication support
- Pipeline stalls while multiplication result is pending
- Dedicated multiplier state keeps forwarded inputs

```
[INFO]: Testing with pattern <Mul>
-----

START!!! Simulation Start .....

-----

=====
Reset ...
=====
sh: ../00_TESTBED/info/success: Permission denied
=====

\\(^o^)/ CONGRATULATIONS!! The simulation result is PASS!!!

=====
$finish called from file "../00_TESTBED/testbench/Final_tb.v", line 269.
$finish at simulation time 718033
```

Performance Patterns

```
[INFO]: Testing with pattern <QSort>
-----
START!!! Simulation Start ....
-----

Reset ...

sh: ../00_TESTBED/info/success: Permission denied

\\(^o^)/ CONGRATULATIONS!! The simulation result is PASS!!!

$finish called from file "../00_TESTBED/testbench/Final_tb.v", line 269.
$finish at simulation time          334831033
```

QSort

```
[INFO]: Testing with pattern <Conv>
-----
START!!! Simulation Start ....
-----

Reset ...

sh: ../00_TESTBED/info/success: Permission denied

\\(^o^)/ CONGRATULATIONS!! The simulation result is PASS!!!

$finish called from file "../00_TESTBED/testbench/Final_tb.v", line 269.
$finish at simulation time          57844033
```

Conv

```
[INFO]: Testing with pattern <LFSR_HIST>
-----
START!!! Simulation Start ....
-----

Reset ...

sh: ../00_TESTBED/info/success: Permission denied

\\(^o^)/ CONGRATULATIONS!! The simulation result is PASS!!!

$finish called from file "../00_TESTBED/testbench/Final_tb.v", line 269.
$finish at simulation time          3250843033
```

LFSR-HIST